IRON METABOLISM

Lecture: Iron Metabolism

Program: BDS

Date: 17/05/16

LEARNING OBJECTIVES

- At the end of the lecture, student should be able to;
  - Know the biochemical properties of iron
  - Enlist the functions of iron in body
  - State the sources of iron in diet
  - Describe the steps of iron absorption from gut to cells
  - Describe its forms of storage in the body
  - Know the clinical importance of iron regulation in body

INTRODUCTION

- Iron is essential to life because of its unusual flexibility to serve as both an electron donor and acceptor
- Vital element in metabolic process
- Deficiency – anemia
- Overload - haemochromatosis



BIOCHEMICAL PROPERTIES OF IRON

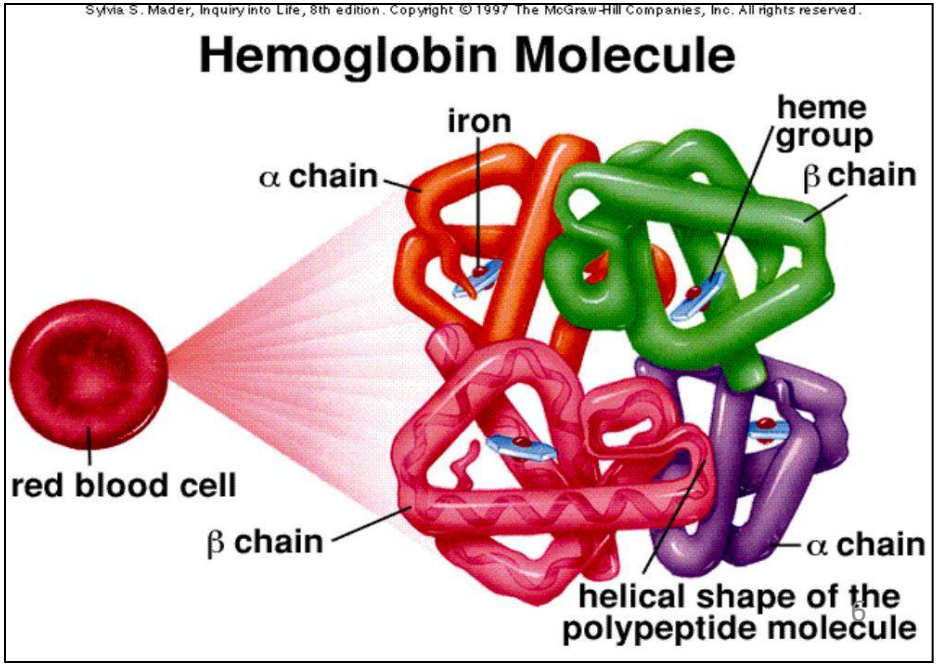
- Iron has an **affinity for electronegative atoms such as oxygen, nitrogen and sulfur,**
  - these atoms are found at the heart of the iron-binding centers of macromolecules.
- In body exist in **two oxidation states**
  - ***ferrous (Fe2+) in acidic medium***
  - ***or ferric (Fe3+) in neutral or alkaline medium***

BIOCHEMICAL PROPERTIES OF IRON

- In the Fe3+ state, iron will form large complexes with anions, water and peroxides.
  - These large complexes have poor solubility and their aggregation leads to pathological consequences.
- Iron also binds to/interfere with the structure and function of various macromolecules.
  - Therefore, the body must protect itself against the deleterious effects of iron. This is the role served by numerous iron-binding proteins

FUNCTIONS

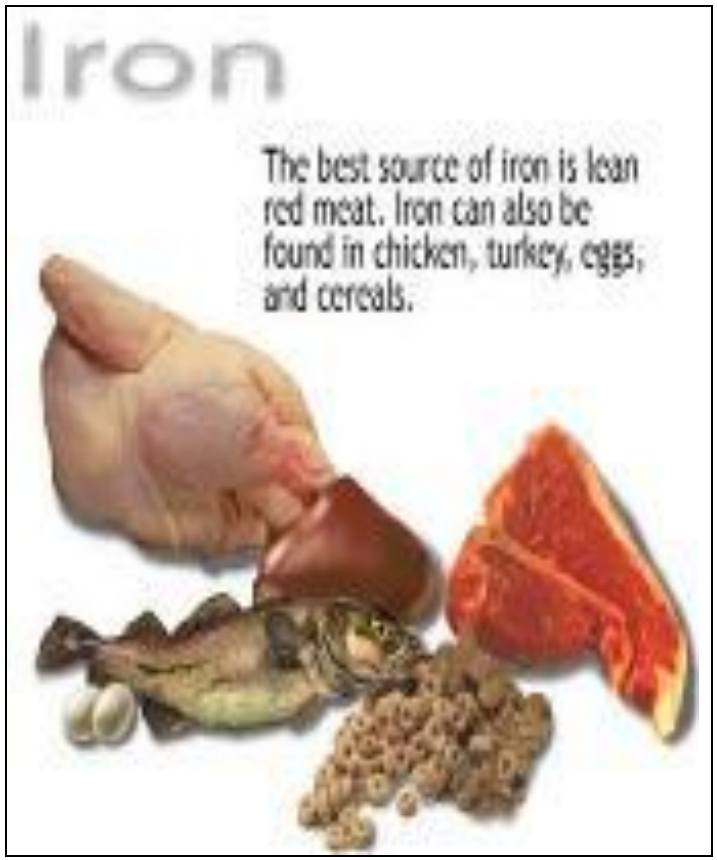
- Iron serves important functions in the body relating to the metabolism of oxygen
  - In hemoglobin: transport O2 & CO2
  - In Myoglobin: reservoir of O2

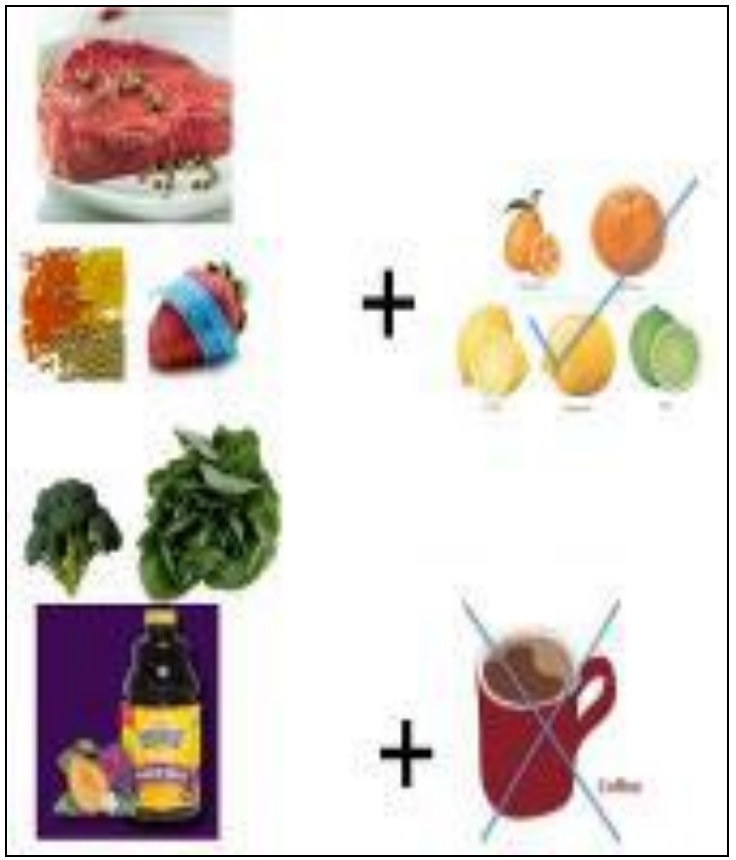


Metabolic functions

- Heme containing enzymes/ETC
  - cytochrome a, b, & c
  - Cytochrome p450
  - Xanthine oxidase
  - Catalase
  - Nitric oxide synthase
  - Prostaglandin synthase (cyclooxygenase)
  - NADH
  - Iron sulphur protein
- Non heme(DNA synthesis & cell differentiation)
  - Ribonucleotide reductase

FOOD SOURCES





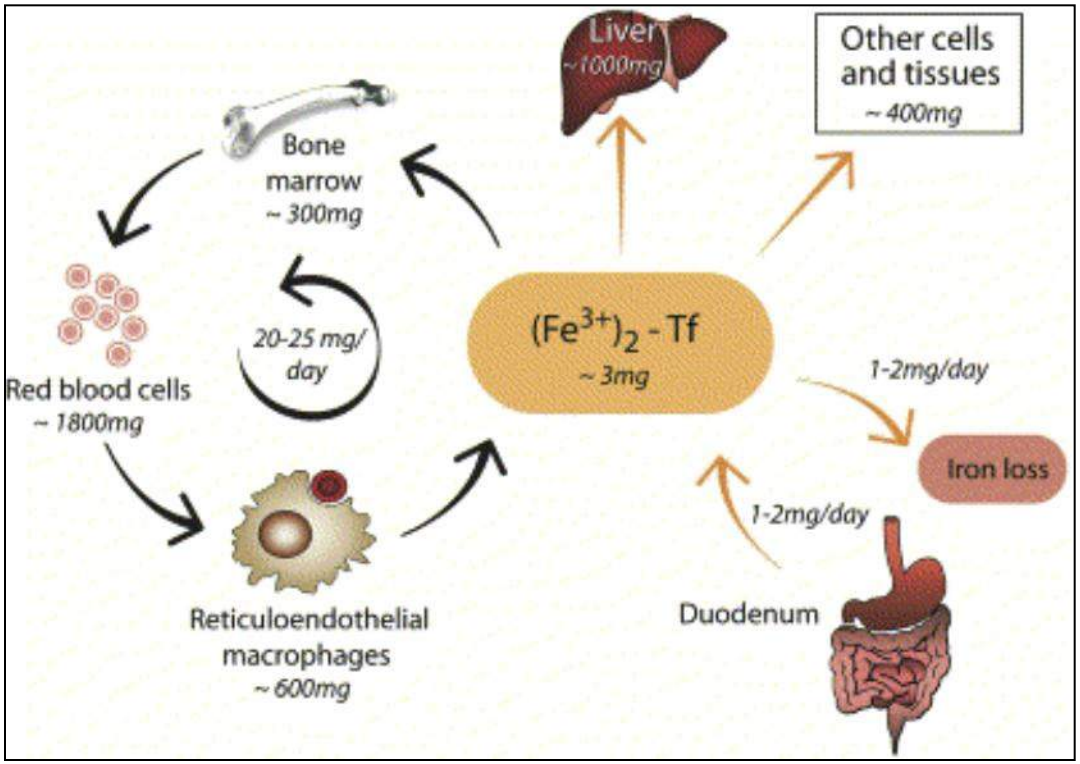
BODY IRON DISTRIBUTION

- 65% of body iron is in haemoglobin
- 30% is in ferritin and haemosiderin
- 3% is in myoglobin
- Erythropoesis occurs in bone marrow
- Iron is transported to bone marrow by plasma transferrin
- Transferrin obtains iron mainly from reticulo-endothelial system (RES)
- Only small proportion comes from dietary iron
- Some iron is stored in RES as ferritin and haemosiderin

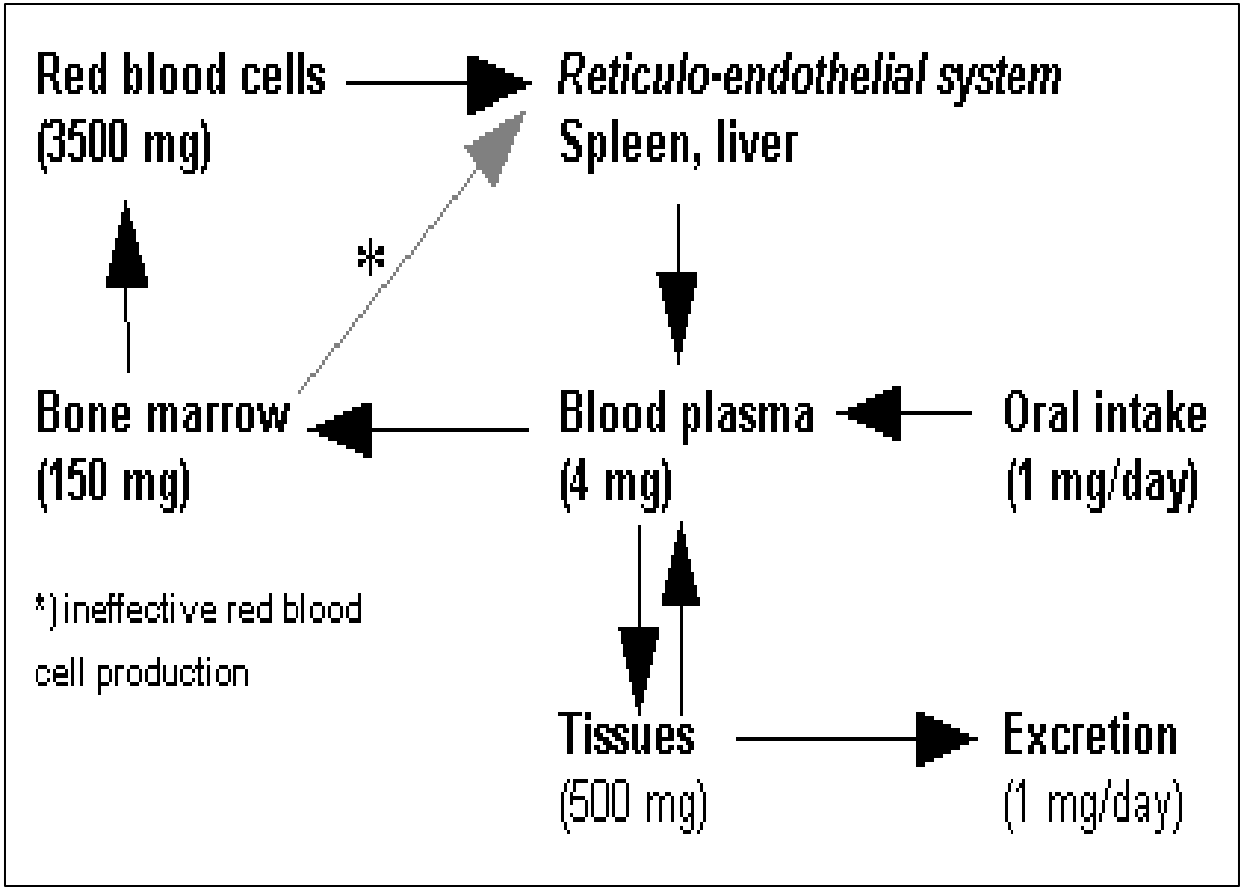
IRON CONTENT OF THE BODY (04 – 05 GRAMS)

- Hemoglobin in RBC 2500/3000 mg
- Myoglobin & Enzymes 500 mg
- Ferritin & Hemosiderin 1000 mg
- Transferrin 4 mg

4004 mg



DISTRIBUTION OF IRON IN BODY

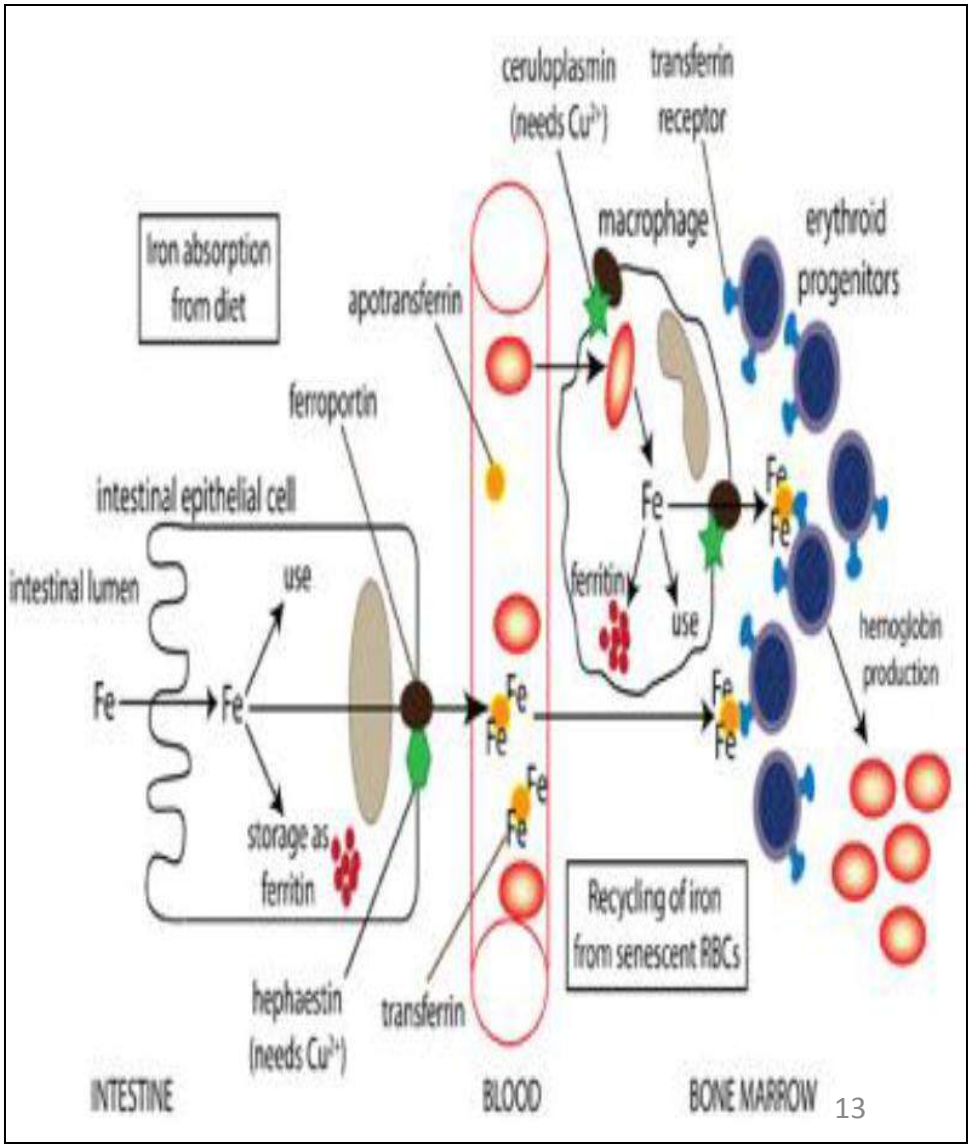


IRON ABSORPTION (1MG/DAY)

- Occur in enterocytes of proximal Duodenum
- All oxidized iron in Fe3+ state is reduced to Fe2+
- Entry of iron is promoted by Divalent Metal Transporter (DMT1).
- Gastric acidity prevents precipitation of insoluble Fe3+
- Hepcidin influence absorption of iron by down regulation.
- Heme is absorbed independent of duodenal pH

IRON UPTAKE

- Inside enterocyte:
  - iron stores as FERRITIN
  - pass across basolateral membrane to be carried to TRANSFERRIN through a protein FERROPORTIN
  - Fe^{2+} is converted to Fe^{3+} by ferroxidase

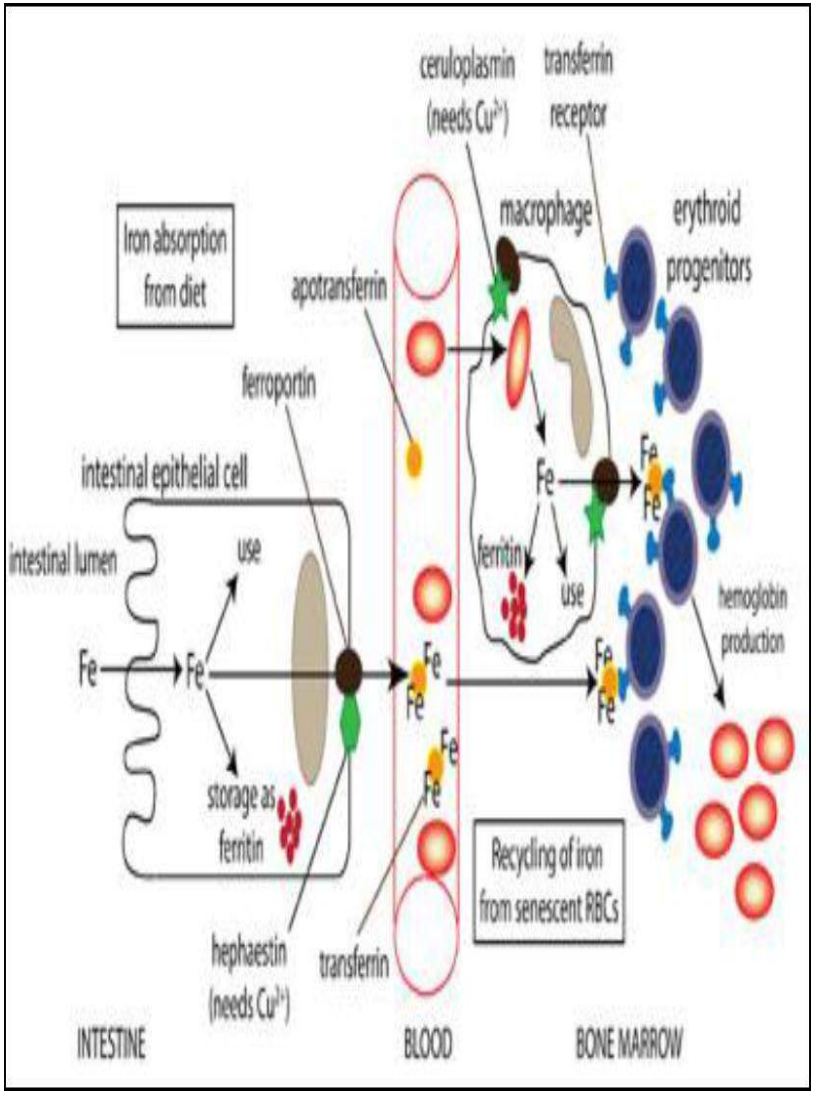


FACTORS AFFECTING ABSORPTION

- Physical State (bioavailability)
  - Heme > Fe^{2+} > Fe^{3+}
- Facilitators
  - Ascorbate, citrate, amino acids, iron deficiency ferric-reductase enzyme on enterocytes.
- Inhibitors
  - Phytates, tannins, soil clay, starch, iron overload, antacids
- Competitors
  - Lead, cobalt, strontium, manganese, zinc

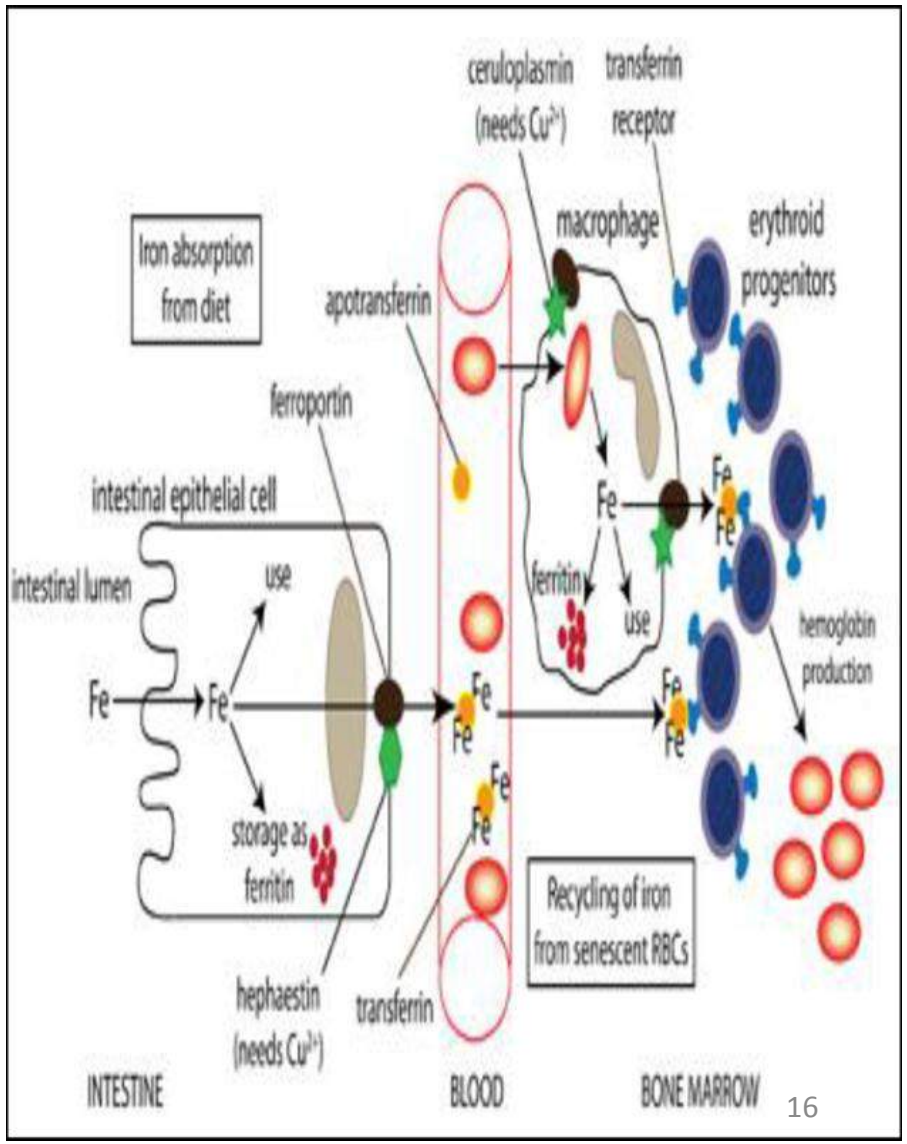
IRON TRANSPORT

- Transported with Transferrin (β1 -globulin)
- Each molecule can binds only 2 Fe^{3+} atoms
- Transport iron from gut to spleen, bone marrow & all cells of body.
- Stored as Ferritin and Hemosiderin.



Storage of iron

- Storage of intracellular iron occurs in liver, skeletal muscle & reticuloendothelial cells.
- Ferritin is the major protein used for intracellular storage of iron.



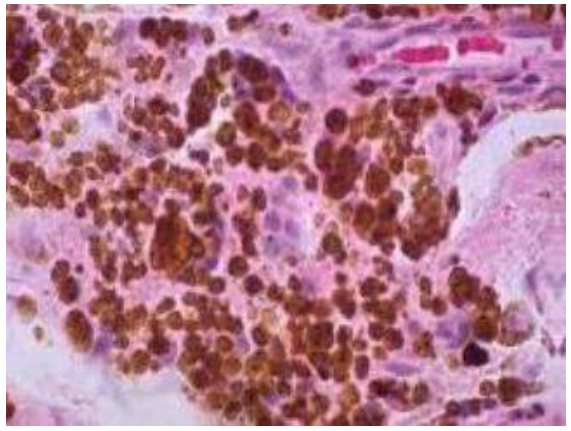
Storage of Iron: FERRITIN AND APO-FERRIITIIN

- In the cell cytoplasm, iron combines mainly with a protein, apoferritin, to form ferritin.
  - Apo-ferritin is a large polymer of 24 polypeptide subunits (binds upto 2,000 iron atoms in the form of ferric-phosphate).
  - Apoferritin has a molecular weight of about 460,000, and varying quantities of iron can combine in clusters of iron radicals with this large molecule
- Therefore, ferritin may contain only a small amount of iron or a large amount.
  - This iron stored as ferritin is called **storage iron**.
- When red blood cells have lived their life span and are destroyed, the hemoglobin released from the cells is ingested by monocyte-macrophage cells. Their, iron is liberated and is stored mainly in the ferritin pool to be used as needed for formation of new hemoglobin.

STORAGE OF IRON (HEMOSIDERIN)

- Hemosiderin is composed of ferritin, denatured ferritin, and other materials and its molecular structure is poorly defined.
- The iron present in hemosiderin is not readily available to the cell and thus, cannot supply iron to the cell when it is needed.





DAILY LOSS OF IRON

- A man excretes about 0.6 milligram of iron each day, mainly into the feces.
- Additional quantities of iron are lost when bleeding occurs.
- For a woman, additional menstrual loss of blood brings long term iron loss to an average of about 1.3 mg/day.

IRON EXCRETION

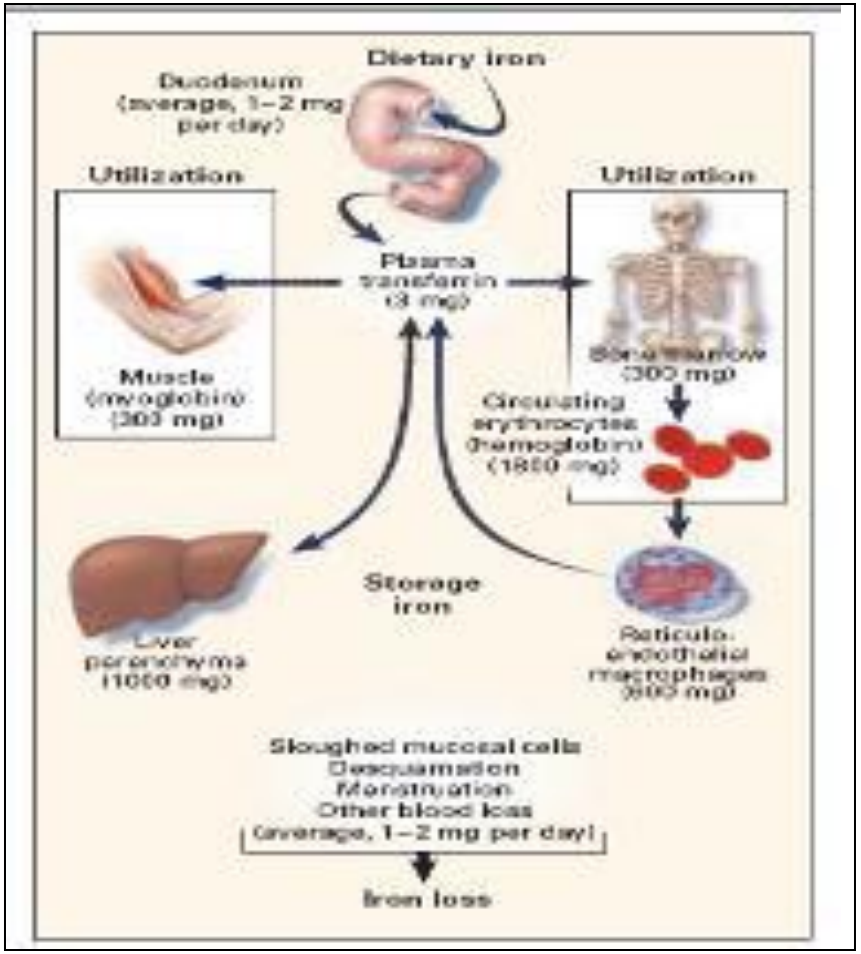
- No physiological pathway to excrete Iron
  - Faeces 80-90% diet
  - Sloughing of GI cell 1-2mg
  - Menstrual flow 5-35mg
  - Pregnancy 350-450mg
  - Accidental bleeding 0.5mg/ml or more

CLINICAL ASPECTS OF ABNORMAL IRON METABOLISM

- HEMOCHROMATOSIS
- IRON DEFICIENCY ANEMIA

IRON METABOLISM, SUMMARY

In the balanced state, 1 to 2 mg of iron enters and leaves the body each day. Dietary iron is absorbed by duodenal enterocytes. It circulates in plasma bound to transferrin. Most of the iron in the body is incorporated into hemoglobin in erythroid precursors and mature red cells. Approximately 10 to 15 percent is present in muscle fibers (in myoglobin) and other tissues (in enzymes and cytochromes). Iron is stored in parenchymal cells of the liver and reticuloendothelial macrophages. These macrophages provide most of the usable iron by degrading hemoglobin in senescent erythrocytes and reloading ferric iron onto transferrin for delivery to cells.



IRON METABOLISM, SUMMARY

- Iron absorbed from the intestine is stored as ferritin in intestinal epithelium or transported in plasma as transferrin.
- Erythroid progenitors obtain iron for hemoglobin synthesis from plasma transferrin or from recycling of senescent erythrocytes by macrophages in bone marrow, spleen and liver.
- Iron that is in excess for that required for hemoglobin production is stored in macrophages as ferritin, which is oxidized to hemosiderin. These stores can be released from macrophages in times of need (increased erythropoiesis).

REFERENCES

- LIPPINCOTT'S
- HARPER'S ILLUSTRATED BIOCHEMISTRY

Thank You